AP PHYSICS 1

Unit 1 • Self Check 1.2 • Topics 1.4 and 1.5

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Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A boat moves straight across a river at 6 m/s relative to the water. The current is 8 m/s. What is the boat's speed relative to the bank?

(A) 14 m/s
(B) 2 m/s
(C) 10 m/s
(D) 48 m/s

Question 2

Rain falls straight down at 8 m/s. You run at 6 m/s. At what speed does the rain hit you, in your reference frame?

(A) 2 m/s
(B) 10 m/s
(C) 14 m/s
(D) 8 m/s

Question 3

A ball is thrown horizontally from the top of a building 45 m tall. How long does it take to reach the ground?

(A) 3 s
(B) 4.5 s
(C) 9 s
(D) 2 s

Question 4

During the flight of a projectile (no air resistance), the horizontal component of its velocity

(A) increases
(B) decreases
(C) stays constant
(D) is zero at the top

Question 5

A projectile is launched at 40 m/s at 30° above the horizontal. What is its maximum height above the launch point?

(A) 20 m
(B) 80 m
(C) 40 m
(D) 10 m

Question 6

A force of 10 N acts at 37° above the horizontal ($\sin 37° = 0.6$). What is its vertical component?

(A) 8 N
(B) 6 N
(C) 10 N
(D) 4 N

Question 7 (Free response)

A plane flies horizontally at 60 m/s at a height of 500 m and releases a package.

(a) How long does the package take to reach the ground?
(b) How far does it travel horizontally?
(c) Where is the plane when the package lands? Explain using reference frames.